Saif F. Pathan

1^{st} Year Computer Science Student

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PROJECTS

Bank of Tanzania Treasury: Bond Auction Predictor | *Python, Pandas, Jupyter, Scikit-learn, Seaborn*

- Compiled and cleaned 140+ real BoT treasury bond auction records spanning 5 years across 7 bond tenors directly from official Bank of Tanzania publications
- Engineered the Oversubscription Ratio feature to quantify investor competition and classified auctions into four demand tiers
- Conducted correlation analysis revealing tenor (r = 0.704) as the dominant yield driver while competition showed near-zero impact (r = -0.023), contradicting developed market theory
- Built a Linear Regression yield predictor achieving R^2 = 0.58 and MAE of 1.47%, enabling forward estimation of auction clearing yields
- Identified Tanzania's full monetary policy cycle from 2021 to 2026 through yield trend analysis across all bond tenors

www.github.com/saifstack/bond-auction-predictor

Monte Carlo Stock Price Simulator | *Python, NumPy, Pandas, Plotly, Streamlit*

- Engineered a Geometric Brownian Motion simulation engine capable of generating up to 10,000 independent price paths across a configurable time horizon of up to 2 trading years
- Derived daily price evolution using the GBM stochastic differential equation incorporating Ito's correction to eliminate systematic return overestimation
- Computed institutional-grade risk metrics including Value at Risk (VaR), Conditional Value at Risk (CVaR), Sharpe Ratio, and cross-sectional percentile bands (P5, P25, P75, P95)
- Built and deployed a fully interactive Streamlit dashboard with real-time parameter controls, fan charts, return CDF visualisation, and probabilistic profit analysis
- Designed a modular architecture separating simulation logic from the UI layer enabling the engine to be imported independently for use in other quantitative workflows

https://saifpathan-montecarlo.streamlit.app/
www.github.com/saifstack/MonteCarlo

TECHNICAL SKILLS

Languages & Libraries: Python, HTML, CSS, JS, SQL, Pandas, NumPy, Matplotlib, Scikit-learn,
Tools & Environments: Jupyter Lab, VS Code, Git, GitHub, Microsoft Excel, HackerRank

Finance & Quantitative: Bond Auction Analysis, Yield Curve Dynamics, Correlation Analysis, Linear Regression, Geometric Brownian Motion, Monte Carlo Simulation, Data Cleaning & Feature Engineering

EDUCATION

Sunway University, Malaysia & Lancaster University, UK BSc (Hons) Computer Science · Dual Degree

Indian School – Dar es Salaam, Tanzania · Maths, Physics, Chemistry, Computer Science, English Literature

INTERESTS

Technology Artificial intelligence · Large language models · Open-source development · Running Local LLMs on personal hardware · Self-hosting & home server development

Quantitative finance · Equity research & idea generation · Fundamental analysis · Securities & emerging market investing · Algorithmic trading